

ChE 312 – Transport Processes II

III. Fundamentals of mass transfer.

III.a. Basic concepts and equations: Diffusion, (bulk) Convection, Convective MT.

- Mass transfer fundamentals.
- Flux vs flow.
- Fick's Law.

III.b. Steady-state diffusion.

- Equimolar counterdiffusion ("EMD") vs. A through stagnant B ("UMD", "unimolecular diffusion").
- Applications; HW08.

III.c. Diffusivities from data, correlation, and theory.

- Diffusion coefficients ("diffusivities")
- Transport properties from "Kinetic theory of gases" - T, P, & MW dependence.
- How to measure diffusivities? And permeabilities?
- Compare diffusivities and permeabilities.

Fundamental definitions to remember !!!

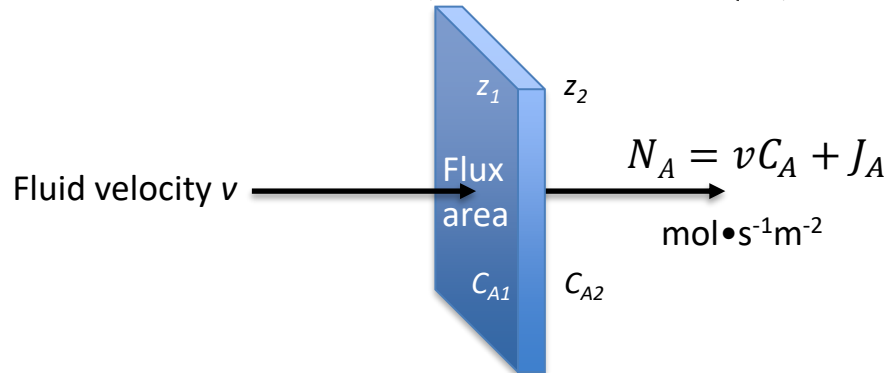
- J_A (diffusion flux) or vC_A (convective flux) or N_A (total flux), mass *or* moles /time per area

- Fick's Law of Diffusion (1855): $J_A = -D_{AB} \frac{dC_A}{dz}$ [Like Fourier's Law, $\frac{q}{A} = -k \frac{dT}{dz}$]

- Bulk convection of species A: vC_A

- Total flux N_A : $N_A = vC_A + J_A \xrightarrow{\text{binary}} = \left(\frac{N_A + N_B}{C} \right) C_A - D_{AB} \frac{dC_A}{dz}$

- [Convective MT to/from a surface: $N_{A,convective} = k \cdot (C_{A,bulk} - C_{A,surface})$]



Where does $N_A = J_A + \frac{(N_A + N_B + \dots)C_A}{C}$ come from?

- Total flux $N_A =$ Diffusion flux $J_A +$ Convective flux vC_A
- Total concentration is C : density in mass/vol or mol/volume, like $C = \frac{N}{V} = \frac{P}{RT}$
- Alternative form: First recognize $N_{total} = [\sum N_i = \sum vC_i = v \sum C_i] = vC$
 - Then $v = \frac{\sum N_i}{C}$ so...
 - Convective flux of A is $vC_A = \frac{\sum N_i}{C} C_A$
 - So $N_A = J_A + \frac{(N_A + N_B + \dots)C_A}{C}$

We will construct and use related forms of two reference cases.

Apply this idea: If N_A and N_B are related, integrate $N_A = J_A + \left(\frac{N_A+N_B}{C}\right) C_A$ by simplifying $\frac{N_A+N_B}{C}$.

1. Equimolar counterdiffusion: If $N_B = -N_A$, then $N_A = J_A + \left(\frac{N_A+N_B}{C}\right) C_A = -D_{AB} \frac{\partial C_A}{\partial z}$

– How can that happen physically? e.g., If $V \leftrightarrow L$ in distillation, or fast reaction $A \rightarrow B$ at a surface.

[Note: If $2A \rightarrow B$ at a surface, then $N_B = -2N_A$ and then $\sum(N_A + N_b) = -N_A$]

– If 1D, constant D_{AB} , & steady-state, then integrate $N_A \int dz = -D_{AB} \int dC_A$ and $N_A \Delta z = -D_{AB} \Delta C_A$

– If C_{A0} at z_1 and C_{A2} at z_2 , then $N_A(z_2 - z_1) = +D_{AB}(C_{A1} - C_{A2})$

– Using $\Delta x_A = C(x_{A1} - x_{A2})$, then

2. Diffusion of A through stagnant B: If $N_B = 0$, then $N_A = J_A + N_A C_A / C$

– Then $N_A = -D_{AB} \frac{dC_A}{dz} + \frac{N_A C_A}{C} = -D_{AB} C \frac{dx_A}{dz} + N_A x_A$ or $(1 - x_A)N_A = -D_{AB} C \frac{dx_A}{dz}$

– Integrate $N_A dz = D_{AB} C \frac{(-dx_A)}{(1-x_A)} = D_{AB} C \cdot d \ln(1 - x_A)$ with

– From $N_A = \frac{D_{AB} C}{z_1 - z_2} \cdot \ln \frac{(1-x_{A2})}{(1-x_{A1})}$, we can get the usefully analogous form

$$N_A = \frac{D_{AB} C}{\Delta z} \Delta x_A$$



$$N_A = \frac{D_{AB} C}{\Delta z \cdot (x_B)_{\log \text{ mean}}} \Delta x_A$$

Manipulate to get $\frac{\Delta x_A}{(x_B)_{\log \text{ mean}}}$ for diffusion of **A** through non-diffusing **B**.

- Recap:

$$N_A = J_A + (N_A + N_B) \frac{C_A}{C} \quad \Rightarrow \quad N_A = -D_{AB} C \frac{dx_A}{dz} + N_A x_A$$

$$N_A (1 - x_A) = -D_{AB} C \frac{dx_A}{dz} \Rightarrow \int_{z_1}^{z_2} dz = -\frac{D_{AB} C}{N_A} \int_{x_{A1}}^{x_{A2}} \frac{dx_A}{1 - x_A}$$

$$N_A = -\frac{D_{AB} C}{z_2 - z_1} \ln \left(\frac{1 - x_{A2}}{1 - x_{A1}} \right) \cdot \frac{x_{A2} - x_{A1}}{x_{A2} - x_{A1}} = \left[\frac{D_{AB} C}{z_2 - z_1} \cdot \frac{\ln[(1 - x_{A2})/(1 - x_{A1})]}{(1 - x_{A2}) - (1 - x_{A1})} \right] (x_{A2} - x_{A1})$$

- Use log-mean definition:

$$f_{\log \text{ mean}} \equiv \frac{f_2 - f_1}{\ln f_2 - \ln f_1} = \frac{f_2 - f_1}{\ln(f_2/f_1)}$$

- To get correction term:

$$\frac{(1 - x_{A2}) - (1 - x_{A1})}{\ln((1 - x_{A2})/(1 - x_{A1}))} \equiv (1 - x_A)_{\log \text{ mean}} = x_{B, \log \text{ mean}} = \frac{x_{B2} - x_{B1}}{\ln(x_{B2}/x_{B1})}$$

$$N_A = \left[\frac{D_{AB} C}{(z - z_1)(x_B)_{\log \text{ mean}}} \right] (x_{A1} - x_{A2})$$

- (1) Same form as $N_A = k_x(x_{A2} - x_{A1})$
- (2) Can use other variables like p_i or C_A etc.

You can do useful things with these relations.

- Suppose you have a vulcanized rubber in a tire where diffusivities at 303K are $D_{O_2,membrane}=2.1E-8 \text{ m}^2/\text{s}$ and $D_{N_2,membrane}=1.5E-8 \text{ m}^2/\text{s}$
- You're told that putting N_2 in your tires will be better than O_2 . Is it because there is less diffusive loss?
- $$N_A = \frac{D_{AB}}{\Delta z} \Delta C_A = \frac{D_{AB}}{\Delta z} \frac{1}{RT} (P_{in}y_{in} - P_{out}y_{out})$$
- Relative loss is
$$\frac{N_{N_2}}{N_{N_2}+N_{O_2}} = \frac{D_{N_2,m}((50\text{psia})(1.00)-(15\text{psia})(0.79))}{D_{N_2,m}((50)(0.79)-(15)(0.79))+D_{O_2,m}((50)(0.21)-(15)(0.21))} =$$

$$\frac{57.225}{41.475+15.435} = \frac{57.225}{56.910} = 1.006$$
- No loss-rate difference at 303 K; Different D at high T? Oxidation by O_2 ?

Theory helps us correlate.

- How does diffusivity change:
 - With T and P ?
 - With molecular weight?
 - With phase: gas, liquid, solid?
- We have a good qualitative intuition.
- But gas correlation is based on theory (see book):

$$D_{AB} \left(\text{m}^2/\text{s} \right) = \frac{1.8583 \times 10^{-7} T^{3/2}}{P \sqrt{S_{AB}} \cdot W_{D,AB}(T)} \left(\frac{1}{M_A} + \frac{1}{M_B} \right)^{1/2}$$

- Extended by correlation (v_A & v_B “specific volumes”; see book)

$$D_{AB} \left(\text{m}^2/\text{s} \right) = \frac{1.00 \times 10^{-7} T^{1.75}}{P \left(\left(\frac{v_A}{\text{cm}^3} \right)^{1/3} + \left(\frac{v_B}{\text{cm}^3} \right)^{1/3} \right)^{1/2}}$$